Cambridge IGCSE Mathematics 0580/22

May/June 2024 - Paper 2 Extended - Duration: 1h 30m - Total Marks: 70

INSTRUCTIONS TO CANDIDATES:
Answer all questions. Use a black or dark blue pen.
You may use an HB pencil for any diagrams or graphs.
Electronic calculators should be used where appropriate.

Question 1 [2 Marks]:
Write 0.0045 in standard form.

Question 2 [7 Marks]:
a Solve 2x^2 + 5x - 12 = 0 by factorisation. [3 marks]
b Solve the simultaneous equations: y = 2x + 1 and x^2 + y^2 = 29. [4 marks]

Question 3 [5 Marks]:
In triangle ABC, AB = 7 cm, BC = 9 cm and angle ABC = 60 degrees.
Calculate the length of AC and the area of the triangle.

Question 4 [3 Marks]:
Find the coordinates of the turning point on the curve y = x^2 - 6x + 11.